PERFORMANCE ANALYSIS OF AERIAL ROBOT MOTORS

A PROJECT REPORT

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ABSTRACT

Unmanned Aerial Vehicles (UAVs) rely heavily on efficient propulsion systems to achieve optimal flight stability and endurance. Among these, Brushless DC (BLDC) motors play a vital role in converting electrical energy into thrust. However, most existing performance data are derived from manufacturer specifications or expensive test systems, which do not reflect real-world operating conditions. This project aims to design and implement a cost-effective experimental setup to analyse the performance of aerial robot motors under varying load conditions. A thrust measurement rig incorporating an ESP32 microcontroller, HX711 load cell amplifier, and Electronic Speed Controllers (ESCs) was developed for real-time data logging and visualization via a web-based dashboard. Controlled experiments were conducted using 5-inch, 7-inch, and 10-inch propellers to assess thrust, current, voltage, and power characteristics. The results revealed that the 7-inch propeller achieved the best balance between thrust and efficiency, while larger propellers increased current draw significantly. The developed system provides a reliable, IoT-enabled platform for UAV propulsion analysis, supporting future research in motor optimization, remote testing, and flight performance prediction.

Keywords: UAV propulsion, BLDC motor, thrust measurement, ESC, ESP32, load cell, motor efficiency

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NOMENCLATURE

ABBREVIATIONS

UAV	Unmanned Aerial Vehicle
ESC	Electronic Speed Controller
ADC	Analog-to-Digital Converter
PWM	Pulse Width Modulation
Li-Po	Lithium Polymer
UI	User Interface
DAQ	Data Acquisition
SD	Secure Digital Card
GUI	Graphical User Interface

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND OF AERIAL ROBOTS

Aerial robots, commonly known as **Unmanned Aerial Vehicles (UAVs)** or **drones**, represent one of the fastest-growing technological domains in modern mechatronics. Their applications extend from military reconnaissance to civilian uses such as agricultural spraying, surveillance, mapping, aerial photography, delivery services, and disaster management. These systems are designed to operate either autonomously or via remote control, depending on their onboard sensors and control algorithms.

The efficiency of an aerial robot is determined by its **propulsion system**, which governs how effectively electrical energy from the battery is converted into mechanical thrust to lift the vehicle. This propulsion subsystem typically consists of **brushless DC (BLDC) motors**, **electronic speed controllers (ESCs)**, **propellers**, and **power management circuits**.

Among these, BLDC motors are the most critical component since they are responsible for generating thrust. They are preferred in UAVs due to their high torque-to-weight ratio, low maintenance requirements, longer lifespan, and superior speed control characteristics. Understanding how these motors behave under varying conditions of load, voltage, and current is vital for efficient UAV design.

With the growing demand for endurance and payload capacity, the **need for performance evaluation of BLDC motors** has become crucial. Aerial robots designed without considering these performance parameters may suffer from instability, overheating, or inefficient energy consumption. Therefore, this project

focuses on **developing a systematic experimental setup** to study motor performance under controlled conditions.

1.2 IMPORTANCE OF PROPULSION SYSTEM IN UAVs

The propulsion system of a UAV converts electrical power into thrust, enabling the vehicle to take off, hover, and maneuver in air. It is the heart of aerial robotics and is directly linked to the vehicle's flight time, payload handling, and overall efficiency.

A typical multi-rotor drone employs multiple propulsion units, each comprising a BLDC motor and propeller, to achieve stability and maneuverability. The selection of an appropriate propulsion system ensures optimal flight performance, energy management, and safety.

Key parameters influencing propulsion system performance include:

- **Thrust-to-weight ratio** – Determines lifting capability.
- **Power-to-thrust ratio** – Indicates energy efficiency.
- **Motor speed (RPM)** – Affects propeller airflow and stability.
- **Current draw and voltage consumption** – Govern flight time and battery life.

Therefore, before integrating motors into UAV platforms, **experimental performance analysis** must be conducted to identify optimal operating points that balance thrust output and energy efficiency.

1.3 PROBLEM STATEMENT

Manufacturers of BLDC motors typically provide specifications such as maximum current, voltage, and thrust under ideal conditions. However, these values often differ when motors are evaluated under real-world environments or

with different propellers and ESCs. Inaccurate performance data can lead to poor design choices, resulting in reduced flight efficiency and motor overheating.

Most existing motor test systems available commercially are **expensive and proprietary**, limiting their use in academic or low-budget research environments. Moreover, they may lack customization options for testing different motors and propellers or real-time dashboards.

Hence, the problem addressed in this project is the **absence of a low-cost, flexible, and accurate test setup** capable of measuring thrust, current, voltage, and efficiency in real time. The study aims to develop such a setup using affordable components like the **ESP32 microcontroller**, **HX711 load cell amplifier**, and **data logging modules**, to experimentally analyze the performance of motors used in aerial robots.

1.4 NEED FOR THE STUDY

The need for this project arises from the increasing importance of **data-driven design** in UAV development. Most drone builders and researchers rely on manufacturer datasheets, which may not account for varying propeller sizes, ESC types, or battery configurations.

This study fulfills several critical needs:

- **To provide empirical data** on the performance of aerial motors under different load conditions.
- **To establish a standardized testing method** for UAV propulsion components.
- **To support educational and research activities** in mechatronics laboratories by creating a reusable testing platform.

- **To integrate IoT capabilities** into traditional motor testing, enabling remote monitoring and digital data storage.

By addressing these needs, the project bridges the gap between theoretical motor specifications and practical performance data, ensuring improved design and decision-making for UAV propulsion systems.

1.5 OBJECTIVES OF THE PROJECT

The main objective of the project is to **analyze the performance of aerial robot motors through an experimental setup** and determine their operational characteristics under varying conditions.

Specific objectives include:

1. To design and fabricate a **motor testing rig** capable of measuring thrust, voltage, current, and power.
2. To integrate sensors and electronic components for precise data acquisition.
3. To interface the system with an **ESP32 microcontroller** for data processing and storage.
4. To develop a **real-time dashboard** for monitoring motor parameters and visualizing performance graphs.
5. To evaluate the motor's efficiency and identify trends that correlate power input with thrust output.
6. To provide practical insights for selecting motors for UAV applications based on experimental results.

1.6 SCOPE AND LIMITATIONS

Scope of the Project:

- The project focuses on **experimental performance analysis** of BLDC motors used in aerial robots under both static and controlled external aerodynamic conditions.
- A **custom thrust test rig** is designed and fabricated to measure thrust, current, voltage, and power parameters of aerial propulsion systems.
- An **additional BLDC motor is incorporated to simulate external wind**, enabling the analysis of motor performance under varying airflow conditions like real flight scenarios.
- The system integrates **sensors, load cell (HX711), and ESP32 microcontroller** for precise data acquisition and logging to a **micro-SD card**.
- A **real-time dashboard interface** is developed for live visualization of thrust, current, voltage, power, and throttle percentage.

Limitations:

- The external wind simulation is restricted to **uniform airflow generation**, which may not fully represent complex aerodynamic flow patterns in actual flight.
- Experiments are conducted under **laboratory conditions**, without considering altitude, humidity, or temperature variations.
- Only one motor–propeller combination is tested at a time.
- The project does not include endurance testing or flight dynamics evaluation; it focuses solely on **bench-level motor performance analysis**.

CHAPTER 2

LITERATURE REVIEW

2.1 TYPES OF MOTORS USED IN UAV PROPULSION

The propulsion system of a UAV is one of the most crucial subsystems, responsible for generating the necessary thrust to lift, hover, and maneuver the aerial platform [1] [2]. The selection of a suitable motor directly influences parameters such as efficiency, payload capacity, flight time, and control stability. Over the years, several types of electric motors have been used in aerial robots, evolving from simple brushed DC motors to highly efficient brushless and coreless designs.

2.1.1 Brushed DC Motors:

Brushed DC motors represent the earliest and most basic form of electric propulsion used in small aerial and electromechanical systems. They operate based on **mechanical commutation**, where a pair of **carbon brushes** and a **commutator ring** periodically reverse the current direction in the rotor winding to produce continuous rotation [5].

The working principle of a brushed DC motor involves the interaction between the **stator magnetic field** (usually generated by permanent magnets) and the **rotor field** (produced by current-carrying coils). The mechanical commutation ensures that torque remains in the same rotational direction. While this design is simple and inexpensive, it introduces several limitations that reduce its suitability for modern UAVs.

Advantages of Brushed DC Motors:

- Simple construction and easy control using direct DC voltage.

- Low initial cost and availability in small sizes.
- High starting torque for basic load movement.

Limitations:

- **High frictional losses** due to physical contact between brushes and commutator.
- Frequent maintenance requirements because of brush wear.
- Limited operational life and reduced efficiency (typically 70–75%).
- Electromagnetic noise and sparking at high speeds.

According to **Rao et al. (2017)**, brushed motors are still used in **toy-grade drones, small ground robots, and educational UAV kits**, where cost and simplicity outweigh performance needs. However, they are **unsuitable for advanced aerial robots** that require high thrust-to-weight ratios, rapid throttle response, and long operational lifespan. Their mechanical commutation system restricts the motor's ability to operate efficiently at high speeds, and the brush wear leads to inconsistent performance over time [2].

For research-grade or industrial UAVs, brushed DC motors have largely been replaced by more advanced alternatives like BLDC and coreless motors that provide higher reliability, efficiency, and control precision.

2.1.2 Brushless DC Motors (BLDC):

Brushless DC motors, commonly known as **BLDC motors**, have become the **standard propulsion system for modern drones and aerial robots** [4] [11] [12]. Unlike brushed motors, BLDC motors use **electronic commutation** through an **Electronic Speed Controller (ESC)** instead of mechanical brushes. This

eliminates frictional contact, reducing mechanical losses and extending motor life.

The BLDC motor consists of a **permanent magnet rotor** and a **stator with multiple coils** arranged in a three-phase configuration. The ESC switches current among the stator coils in a precise sequence, generating a rotating magnetic field that interacts with the rotor's field to produce torque. This process ensures smooth rotation, efficient power conversion, and precise speed control.

Key Characteristics of BLDC Motors:

- **High Efficiency:** Can achieve efficiencies above 85–90% under optimal load conditions.
- **High Power-to-Weight Ratio:** Provides substantial thrust output while maintaining a compact and lightweight structure.
- **Low Maintenance:** Absence of brushes eliminates wear and sparking, enhancing durability.
- **Precise Control:** Electronic commutation allows for accurate control of speed, torque, and direction through PWM signals.

Research by **Kumar and Lee (2019)** demonstrated that BLDC motors deliver superior performance when operated within their **rated voltage and load limits** [12]. The study indicated that operating the motor at 60–80% of maximum throttle provides the best balance between thrust and efficiency. Beyond this range, current consumption increases rapidly, causing heat buildup and efficiency loss.

BLDC motors also exhibit **excellent torque linearity**, meaning the torque output is nearly proportional to the input current. This characteristic simplifies control algorithms used in UAV flight controllers. Moreover, the **three-phase winding**

configuration ensures smooth torque delivery and minimal vibration, which is vital for maintaining onboard sensor stability during flight.

Another notable advantage is the **compatibility of BLDC motors with variable-pitch propellers** and regenerative braking systems, making them versatile for various aerial robotics applications.

2.1.3 Coreless DC Motors:

Coreless DC motors are compact, high-speed variants of conventional DC motors, primarily used in **miniature UAVs (Micro Aerial Vehicles – MAVs)**, **robots**, and **precision instruments**. In these motors, the rotor winding is wound without an iron core, typically on a thin cylindrical structure supported by the motor shaft. This unique design reduces the rotor's inertia drastically, allowing extremely fast acceleration and deceleration.

The absence of an iron core minimizes magnetic losses and prevents cogging torque, resulting in **exceptionally smooth rotation** even at very low speeds. Due to their light construction, coreless motors are ideal for applications requiring **fast response** and **low inertia**, such as small drones and gimbal stabilization systems.

Advantages of Coreless Motors:

- **Low Inertia:** Enables quick speed changes and precise control.
- **Compact and Lightweight:** Suitable for weight-sensitive designs.
- **Smooth Operation:** Absence of cogging torque ensures stable rotation.

Disadvantages:

- **Thermal Limitation:** Lack of a metallic core reduces heat dissipation, leading to overheating at high speeds.

- **Lower Power Output:** Not suitable for heavy-lift or high-thrust applications.
- **Limited Efficiency:** Efficiency decreases with prolonged high-load operation due to coil deformation risks.

Kwon et al. (2020) studied the performance of coreless DC motors in small aerial vehicles and found that although these motors achieve faster rotational speeds than equivalent BLDCs, their efficiency drops sharply under continuous load. Heat accumulation within the winding limits their operational duration, making them ideal for **short-duration flights or stabilization tasks**, rather than propulsion in large UAVs [2] [11].

Furthermore, their **small form factor** and **low torque output** restrict their use to **micro drones**, **nano UAVs**, and **light payload applications**, where maneuverability is prioritized over endurance. Despite these limitations, coreless motors remain popular for experimental UAVs and academic research platforms due to their low cost and simplicity of control.

2.2 COMPARISON OF POWER-TO-WEIGHT RATIO AND EFFICIENCY

The **Power-to-Weight Ratio (PWR)** is one of the most fundamental performance indicators in aerial robotics. It quantifies the **amount of thrust a propulsion system can produce relative to its total weight**, which includes the motor, propeller, and electronic control components. A higher PWR implies that the aerial vehicle can lift more payload, accelerate faster, and perform agile maneuvers. For unmanned aerial vehicles (UAVs), especially multirotors, maintaining an optimal power-to-weight ratio is critical to achieving stable and efficient flight [1] [3].

In UAV design, thrust should typically be at least **twice the total weight of the vehicle** to ensure adequate maneuvering capability and stability during takeoff, hovering, and descent. If the thrust produced by the propulsion system is barely equal to the UAV's weight, the motors will operate at near-maximum throttle, leading to excessive heating and shortened lifespan. Thus, selecting a propulsion system with an appropriate PWR is essential for balancing **endurance, payload capacity, and motor reliability** [11].

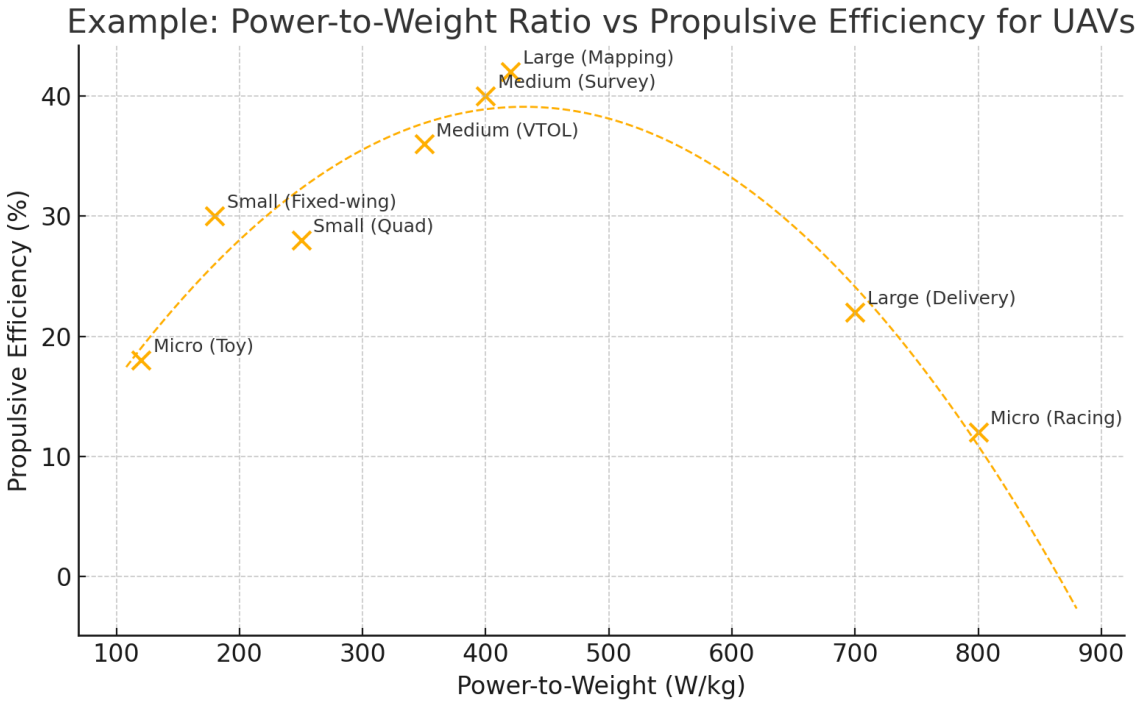


Figure 2.1 Power-to-Weight ratio vs Propulsive Efficiency

Zhou et al. (2018) conducted a comprehensive comparison of UAV propulsion systems and found that **Brushless DC (BLDC) motors** consistently outperformed **brushed** and **coreless** motors in both efficiency and sustained thrust generation. Their experimental analysis revealed that BLDC motors delivered up to **30–40% higher thrust-to-weight ratios** due to their superior torque characteristics and minimal mechanical losses [3]. However, the researchers also noted that efficiency decreases at higher throttle levels because of increased current draw and resistive heating in motor windings. As throttle

increases, aerodynamic drag on the propeller rises sharply, demanding more power from the motor and thus reducing the overall system efficiency.

Further studies by **Singh and Patel (2021)** highlighted that UAV propulsion efficiency is not determined by the motor alone but by the **combined performance** of the motor, **Electronic Speed Controller (ESC)**, and **propeller design**. They demonstrated that even a highly efficient motor could perform poorly if paired with an incompatible ESC or propeller [13]. For instance, using a propeller with a pitch angle unsuitable for the motor's speed constant can cause the motor to draw excessive current, reducing flight time and increasing thermal stress.

The researchers categorized propulsion losses into three major types [3] [11]:

1. **Electrical Losses** – Occur due to resistance in motor windings and inefficiencies in ESC switching.
2. **Mechanical Losses** – Result from friction in bearings and aerodynamic drag on the rotor.
3. **Aerodynamic Losses** – Associated with propeller slipstream effects and turbulent airflow, especially at high RPMs.

They concluded that the **synergistic tuning** of motor, ESC, and propeller can improve UAV propulsion efficiency by 10–15% compared to random component combinations. Hence, the **experimental evaluation** of each motor–propeller–ESC combination is crucial to optimize power utilization and extend flight endurance.

From the literature, it is evident that a **high power-to-weight ratio combined with optimal energy conversion efficiency** is the cornerstone of UAV propulsion design. The present project addresses this requirement by

experimentally determining thrust, voltage, current, and power under various operating conditions. By analyzing these parameters, it is possible to derive efficiency curves and identify the most effective operating points for aerial robot motors, ensuring longer flight duration and reliable performance under both static and aerodynamic conditions.

2.3 THRUST–POWER CHARACTERISTICS OF UAV MOTORS

The relationship between **thrust** and **power consumption** forms the foundation of UAV propulsion design. Thrust determines the **lifting capability** of the aerial vehicle, whereas power governs the **energy efficiency and endurance** of the system. For an aerial robot to achieve stable and efficient flight, it must generate adequate thrust to overcome its total weight while minimizing power consumption to extend flight time [3] [4] [11].

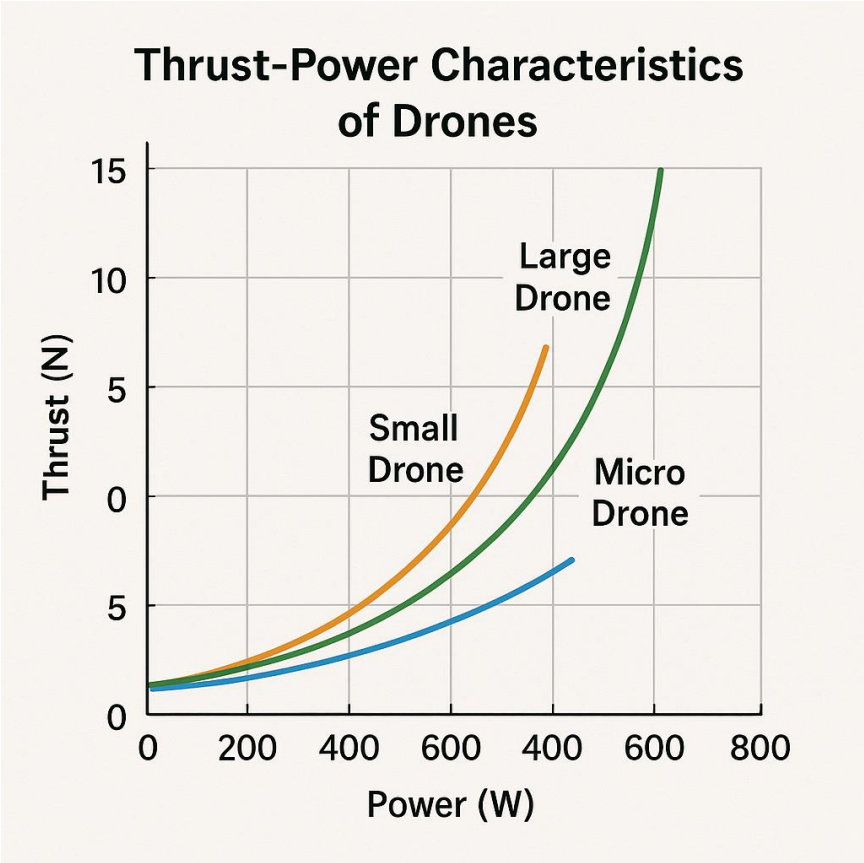


Figure 2.2 Thrust-Power Characteristics

Fundamental Relationship Between Thrust and Power

In aerodynamic propulsion, thrust (T) is primarily a function of the **rotational speed** of the propeller and the characteristics of the airflow it generates. For a given propeller design, thrust is approximately proportional to the **square of the propeller's rotational speed (n)**, while power consumption (P) increases roughly with the **cube of the rotational speed**. Mathematically, these relationships can be expressed as:

$$T \propto n^2 \text{ and } P \propto n^3$$

This means that even a slight increase in propeller speed results in a significant rise in power demand. Consequently, UAV designers face a critical trade-off between **increasing thrust output** and **maintaining power efficiency**.

For example, doubling the motor speed theoretically increases thrust by four times but increases power consumption by eight times. This exponential relationship implies that small throttle adjustments can drastically affect battery drain and thermal performance.

2.4 ELECTRONIC SPEED CONTROLLER (ESC) AND MOTOR INTERACTION

The **Electronic Speed Controller (ESC)** is one of the most vital components in the propulsion system of Unmanned Aerial Vehicles (UAVs). It acts as the **interface between the power source (battery)** and the **motor**, regulating the amount of current supplied based on control inputs. The ESC governs the **speed, direction, and braking** of the motor by generating high-frequency **Pulse Width Modulated (PWM)** signals, which effectively control the power delivered to the motor windings.

In UAVs using **Brushless DC (BLDC)** motors, the ESC performs the role of **electronic commutation**, sequentially energizing the stator windings to maintain continuous rotation of the rotor. This replaces the mechanical commutation found in brushed motors, thereby eliminating frictional losses and improving reliability [5] [11].

Operating Principle of ESCs

An ESC primarily converts the **DC voltage from the Li-Po battery** into a **three-phase AC signal** suitable for driving BLDC motors. It receives throttle commands from a **microcontroller or flight controller** in the form of PWM signals. These signals define the motor speed, typically within a duty cycle range of 1 ms (minimum throttle) to 2 ms (maximum throttle) at a frequency of around 50–400 Hz.

The ESC's **microprocessor** interprets these commands and switches **MOSFET transistors** at high frequencies (typically between 8 kHz and 32 kHz) to modulate the output voltage and current supplied to the motor's three phases. This switching frequency determines how smoothly the motor runs and how efficiently electrical power is converted into mechanical energy.

Higher switching frequencies reduce torque ripple and audible noise but also increase switching losses and heat generation. Therefore, the ESC must be designed and calibrated carefully to balance **efficiency, response time, and thermal stability** [11].

Role of ESC in UAV Propulsion

In UAV systems, the ESC performs several critical functions beyond simple motor speed control. These include:

1. **Throttle Signal Interpretation:** Converting PWM signals from the flight controller into precise voltage outputs.
2. **Back-EMF Detection:** Identifying the position of the rotor to ensure correct commutation timing.
3. **Current Regulation:** Preventing excessive current draw to protect the motor and battery.
4. **Thermal Protection:** Monitoring temperature to prevent overheating during extended operation.
5. **Regenerative Braking:** In advanced ESCs, energy can be partially recovered during deceleration phases.

Because each UAV motor has unique electrical characteristics (such as winding resistance, inductance, and Kv rating), the ESC must be matched carefully to the motor's specifications. Using an incompatible ESC may result in **poor linearity, voltage spikes, or unstable motor behaviour**, directly affecting flight performance and safety.

2.5 GAPS IDENTIFIED IN PREVIOUS RESEARCH

From the extensive literature review on UAV propulsion systems, it is evident that significant progress has been made in understanding motor performance, thrust–power characteristics, and propeller aerodynamics. However, most of the available research focuses primarily on **theoretical modelling** or **high-cost experimental test stands** [3] [4] [6] [13]. Despite advancements in UAV design and control, several gaps still persist in the experimental analysis of aerial robot motors.

The key research gaps identified are:

1. Lack of low-cost, modular UAV motor test benches for academic and research applications.
2. Limited experimental analysis under simulated aerodynamic (external wind) conditions.
3. Absence of real-time data visualization and automated performance logging.
4. Insufficient correlation between electrical input and mechanical output parameters.
5. Limited studies on ESC–motor synchronization and efficiency effects.
6. Minimal integration of IoT-based or wireless monitoring systems in propulsion analysis.

Addressing these gaps is essential to improve understanding of UAV propulsion efficiency and to develop more practical tools for motor evaluation. The present project aims to bridge these gaps by designing a **cost-effective, sensor-integrated experimental setup** with **real-time dashboard visualization** and **aerodynamic wind simulation**, offering a more comprehensive performance analysis framework for aerial robot motors [8] [15].

CHAPTER 3

MATERIALS AND METHODS

This chapter details the model's technical architecture, including design specifications, circuit connectivity, and implementation methodology.

3.1 ELECTRICAL SCHEMATIC DIAGRAM

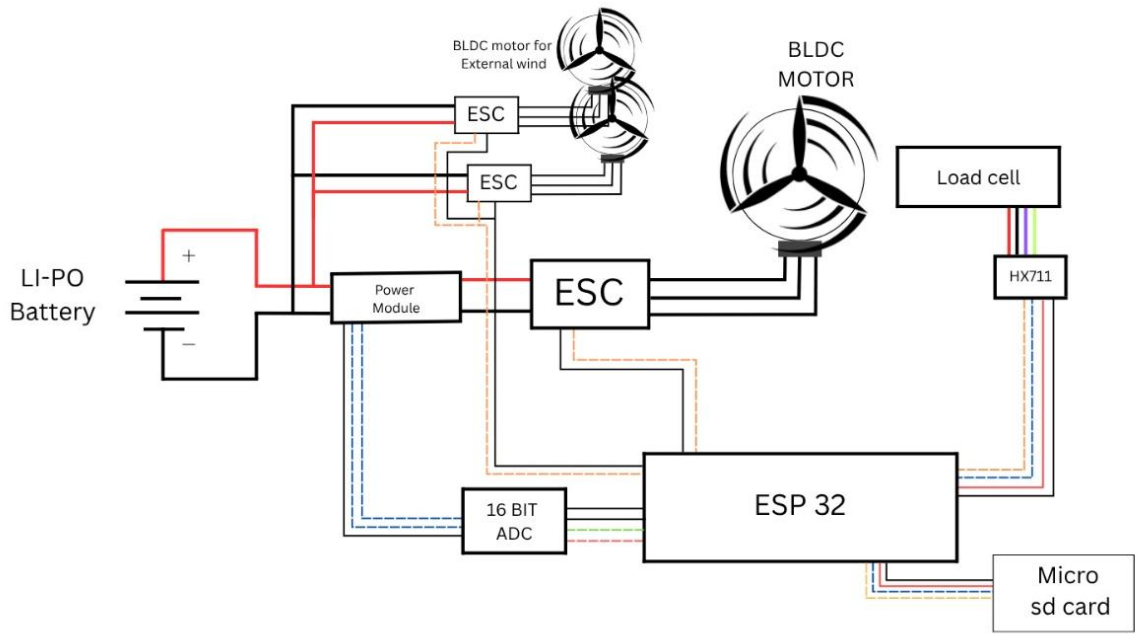


Figure 3.1 Electrical Schematic Diagram

The **Li-Po battery** serves as the primary power source, supplying regulated DC voltage to all components through the **power module**. The power module distributes power to the ESCs and sensors while also providing current and voltage monitoring to protect the system from overload or voltage drops.

Three **Electronic Speed Controllers (ESCs)** are used in the setup. One drives the **main BLDC motor** (under test), another controls the **secondary BLDC motors** that generates external airflow. Each ESC receives PWM control signals

from the **ESP32 microcontroller**, which adjusts the motor speed according to the programmed throttle percentage.

The **main BLDC motor** generates thrust through its propeller, which is measured using a **load cell** mounted behind it. The load cell's small analog output is amplified and converted to a digital signal by the **HX711 module**, which transmits data to the ESP32 for processing.

A **16-bit ADC** is used to accurately measure voltage and current from the power module. These readings are sent to the ESP32, which calculates electrical power and efficiency during each test. The processed data are then saved on a **micro-SD card** for later analysis and also transmitted to a **real-time dashboard** via Wi-Fi.

In summary, the electrical connections ensure smooth power delivery, precise control of motor speed, and accurate acquisition of thrust and electrical data. The setup provides a reliable platform for analyzing the performance of aerial robot motors under both static and wind-influenced conditions.

3.2 DESIGN OF THE SYSTEM

The designed test rig enables precise measurement of thrust, power consumption, and efficiency under varying operating conditions. The system primarily consists of a BLDC motor, a propeller, a load cell, an Electronic Speed Controller (ESC), and a structural frame to ensure stability and repeatability during testing.

The BLDC motor is mounted securely on an aluminium thrust stand, which is designed to minimize vibrations and allow accurate force transmission to the load cell. The load cell is positioned behind the motor mount to sense the thrust generated by the rotating propeller. The ESC is connected to the motor and controlled through a microcontroller (ESP32), allowing variable throttle input to test performance across different speeds.

Figure 3.2 Design of Test Stand

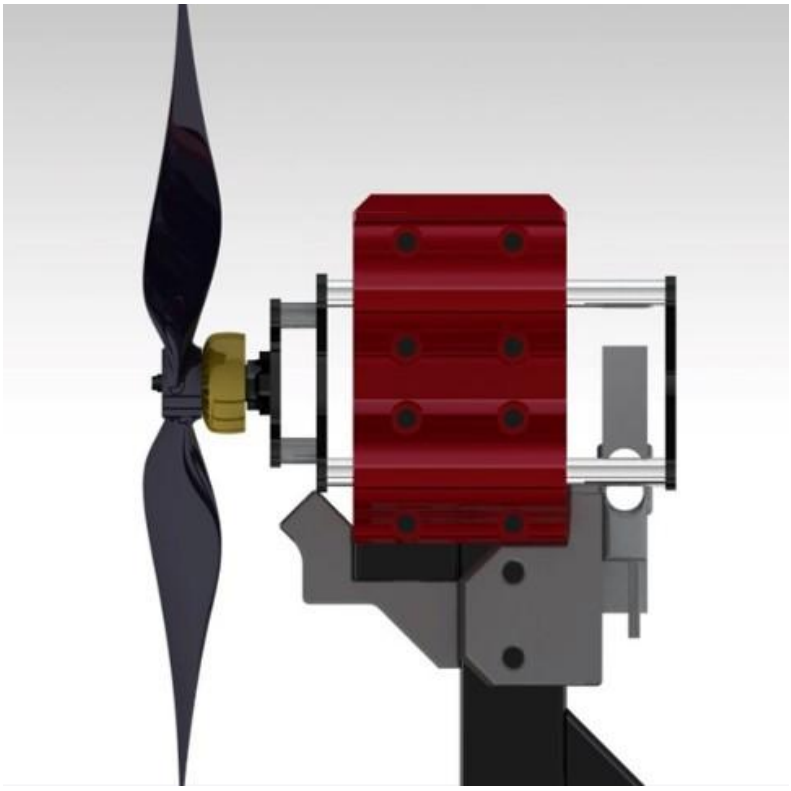


Figure 3.3 Detailed View of Motor Mount

The entire setup is constructed using aluminium extrusion frames to ensure rigidity and modularity. The vertical and horizontal members support the thrust stand and provide mounting points for electronic components such as the power supply and data acquisition system. The propeller is mounted in front of the motor and rotates freely to generate upward thrust, which is measured as a downward reaction force on the load cell. The test rig also includes a power analyzer to measure input voltage and current, enabling computation of power-to-thrust ratio and efficiency.

This rig design allows detailed analysis of drone propulsion performance by correlating motor input parameters with output thrust. It serves as an essential platform for validating motor-propeller combinations, optimizing drone configurations, and conducting aerodynamic testing under controlled conditions.

3.3 COMPONENTS USED

The thrust measurement test rig comprises multiple integrated subsystems, each designed to perform a specific function for evaluating drone motor performance. These subsystems include the **power delivery unit**, **control and data acquisition module**, **thrust sensing setup**, and **structural framework**. Together, they ensure precise measurement of **thrust, voltage, current, and efficiency** under various operating conditions.

The setup is designed to provide stable mechanical support and accurate sensor alignment, minimizing vibration and measurement error. The control and data acquisition modules, managed by the **ESP32 microcontroller**, enable real-time monitoring and data logging for detailed performance analysis. The modular nature of the rig allows easy modification for testing different motors or propellers, making it suitable for research and academic applications in UAV propulsion studies.

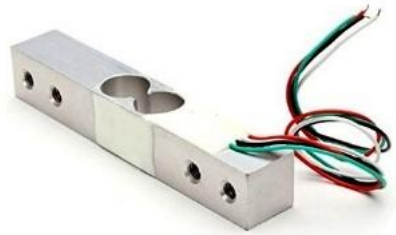
COMPONENT	QUANTITY
Electronic Speed Controllers (ESCs)	3
Power Module	1
HX711 - Load Cell Amplifier	1
ESP32	1
16-bit ADC Module	1

Table 3.1 Components Utilized

COMPONENT	FUNCTIONAL DESCRIPTION
Li-Po Battery	Provides DC power supply to ESCs and control circuit.
Power Module	Monitors voltage and current; provides power to ESP32 and ADC.
Electronic Speed Controllers (ESCs)	Regulates motor speed using PWM signals from ESP32.
Load Cell	Measures thrust force generated by the motor.
HX711 Load Cell Amplifier	Amplifies and digitizes load cell signal for ESP32 input.
ESP32	Controls ESCs, reads sensors, and logs experimental data.
16-bit ADC Module	Converts analog signals (voltage/current) into digital data.
Aluminium Frame Structure	Provides rigid support for mounting motor, sensors, and electronics.

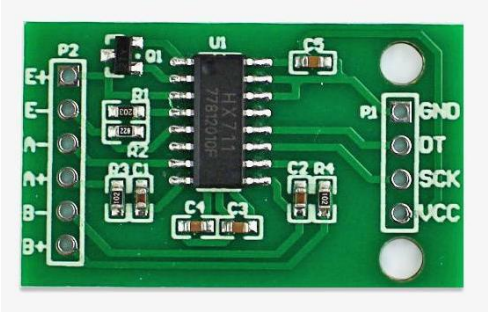
Table 3.2 Functional Description of Key Components

BLDC Motor

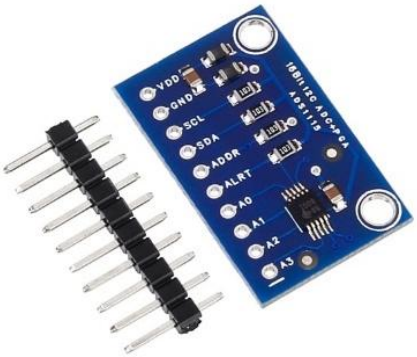


Load Cell (10kg)

ESC



HX711 Amplifier



16-bit ADC

ESP32 Microcontroller

Figure 3.4 Components utilized

3.3.1 Brushless DC Motors

The **BLDC motors** used in this experiment are specifically designed for UAV propulsion. These motors provide high torque-to-weight ratios, excellent speed control, and long operational life. The motors operate efficiently over a wide voltage range (8–15 V) and are capable of delivering thrust sufficient for small and medium UAV platforms.

Two motors were used:

- **Primary Motor:** Mounted on the thrust measuring stand to analyze performance.
- **Secondary Motors (Wind Generator):** Placed laterally to simulate external airflow.

3.3.2 Electronic Speed Controller (ESC)

The ESC regulates the speed and direction of the BLDC motors by controlling the supplied current based on input PWM signals from the ESP32 microcontroller. It acts as the interface between the control system and the motors, ensuring smooth and responsive speed variations. Each motor uses an independent ESC equipped with a Battery Elimination Circuit (BEC) to power the control electronics. Compact and efficient, ESCs are essential for precise motor control and stable operation under varying load conditions.

Load Cell and HX711 Module

The load cell measures the thrust produced by the motor using strain gauge sensing, where mechanical deformation is converted into a proportional electrical signal. A 10 kg single-point load cell is used for its high sensitivity and linearity in low-force applications. The weak analog signal from the load cell is amplified and digitized by the **HX711 module**, a 24-bit precision ADC, before being

processed by the ESP32. This ensures accurate thrust measurement with minimal noise and high resolution.

3.3.3 ESP32 Microcontroller

The ESP32 serves as the central control and data acquisition unit. It collects sensor data from the HX711, voltage and current sensors, and tachometer, then computes key parameters such as thrust, power, and efficiency. The processed data is stored on a micro-SD card and simultaneously displayed on a web-based dashboard via the ESP32's built-in Wi-Fi. With its dual-core processor (240 MHz), low power consumption, and extensive I/O support, the ESP32 enables efficient real-time monitoring and analysis of the propulsion system's performance.

The circuit connection is shown below:

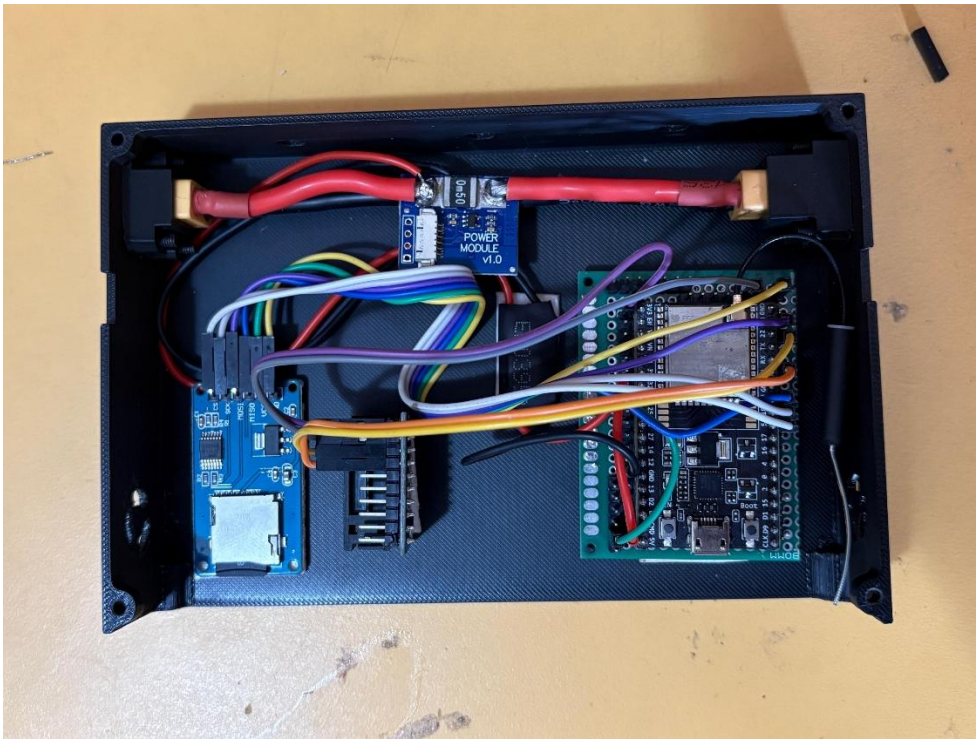


Figure 3.5 Circuit Connection

3.3.4 External Wind Generator

To simulate aerodynamic conditions, a **secondary BLDC motor with a ducted propeller** was positioned facing the primary motor. This external motor generates a controlled wind flow across the propeller, simulating real flight airflow conditions. The wind speed was varied by adjusting the secondary motor's throttle.

This setup allows the study of motor performance under **different aerodynamic load conditions**, such as headwind and crosswind effects, which are typically experienced by UAVs during outdoor operation.

3.4 EXPERIMENTAL PROCEDURE

Mounting and Setup

1. The BLDC motor was securely mounted on the thrust rig with the propeller facing upward.
2. The load cell was calibrated using known weights to ensure accurate force measurement.
3. The ESC was connected to the motor, power supply, and signal pin of the ESP32 microcontroller.
4. Current and voltage sensors were integrated into the circuit for simultaneous measurement.
5. The secondary wind motor was positioned at a distance from the main propeller to generate airflow.

Data Acquisition and Logging

1. The microcontroller was programmed to continuously collect sensor data at a sampling rate set using the dashboard.

2. Parameters such as **thrust (N)**, **voltage (V)**, **current (A)**, **power (W)**, and **efficiency (%)** were measured and displayed in real time on the monitoring dashboard.

3. The motor was operated at gradually increasing throttle levels (20%, 40%, 60%, 80%, and 100%), and corresponding readings were recorded at each stage.

4. The process was repeated with the **wind generator active** to study aerodynamic effects.

5. The experimental readings are logged and can be exported as a **CSV file.**

3.5 EXPERIMENTAL SETUP

The experimental setup is designed to measure the thrust and electrical performance of a BLDC motor. It integrates both mechanical and electrical subsystems to provide reliable and repeatable results. The main components include the BLDC motor with propeller, thrust measurement stand, load cell with HX711 amplifier, ESCs, ESP32 microcontroller, power supply system, and data logging module.

The BLDC motor is mounted on a rigid aluminium extrusion frame using a custom-designed holder to minimize vibration and ensure stable operation. The propeller is attached to the motor shaft, generating a thrust force when powered. The resulting reaction force is transmitted to the load cell, which converts the mechanical force into an electrical signal proportional to the thrust produced. The HX711 amplifier conditions and digitizes this signal before sending it to the ESP32 microcontroller.

The power to the motor is supplied by a Li-Po battery through an Electronic Speed Controller (ESC), which receives a PWM signal from the ESP32 to vary the motor speed. Additional sensors and modules, including a 16-bit ADC and power

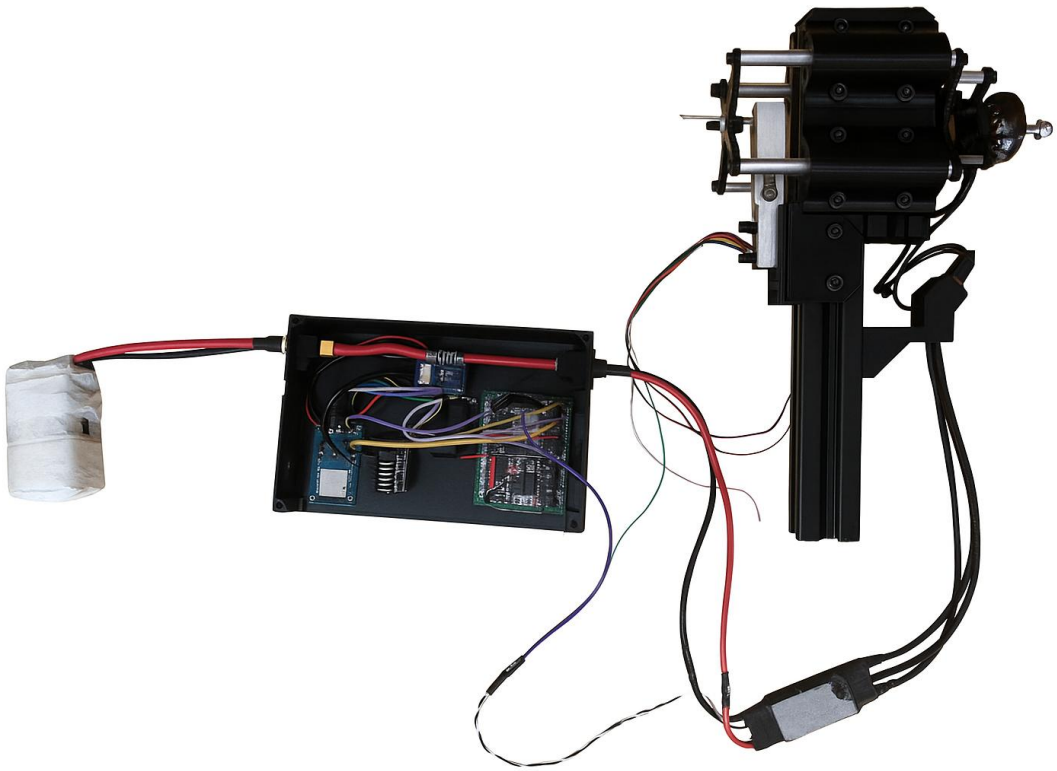


Figure 3.6 Connection of Sensors to Motor

module monitoring voltage and current drawn by the motor for efficiency calculation. The ESP32 processes all incoming data and stores it on a micro-SD card while simultaneously displaying real-time parameters through a Wi-Fi-based dashboard.

The entire setup is designed to ensure structural rigidity, measurement accuracy, and operational safety. All components are securely mounted on an aluminium extrusion frame to minimize vibration and mechanical noise during operation. The wiring layout between the ESCs, sensors, and microcontroller is neatly routed to reduce electromagnetic interference and maintain signal integrity. The integration of the mechanical and electrical subsystems allows synchronized operation between motor control, thrust sensing, and data acquisition. Proper grounding and isolation are provided to prevent power fluctuations from affecting sensor accuracy. Overall, the setup ensures reliable, repeatable, and precise

measurement of thrust and electrical performance, forming a robust foundation for UAV motor characterization.

Figure 3.7 Experimental setup of Thrust measurement rig

This experimental setup enables the measurement of thrust, current, voltage, and power at different throttle levels, allowing the evaluation of motor efficiency and performance characteristics essential for UAV propulsion system analysis.

CHAPTER 4

DESIGN AND DEVELOPMENT OF DASHBOARD

4.1 INTRODUCTION

The thrust measurement system includes a web-based dashboard developed for real-time monitoring, control, and data visualization of motor performance. The dashboard serves as the user interface for interacting with the test rig, allowing users to set throttle levels, initiate thrust tests, and visualize live parameters such as voltage, current, thrust, power, and efficiency. It provides an intuitive control environment, enabling researchers to analyze drone propulsion performance without manual intervention.

4.2 SYSTEM ARCHITECTURE

The system architecture integrates both hardware and software components to ensure efficient data acquisition, processing, and visualization. The ESP32 microcontroller acts as the central control unit, responsible for reading sensor inputs, processing the acquired data, and transmitting it wirelessly to the dashboard via Wi-Fi. The sensors include a load cell (connected through the HX711 amplifier) for thrust measurement and voltage/current sensors for electrical performance monitoring.

The hardware system is powered by a Li-Po battery, with Electronic Speed Controllers (ESCs) regulating the speed of the BLDC motors based on PWM signals from the ESP32. All measurement and control signals are processed digitally to minimize noise and improve accuracy. The processed data is then displayed in real-time on the web-based dashboard, which allows users to control the motor throttle and visualize system response instantly.

The dashboard is hosted locally on the ESP32 web server, enabling users to access it from any Wi-Fi-enabled device such as a laptop or smartphone. The Micro SD card module is integrated for continuous data logging, ensuring that experimental data can be stored and analyzed later for performance evaluation and calibration.

Key modules include:

- **Data Acquisition Unit:** ESP32, ADC, and HX711 for capturing sensor data with high precision.
- **Processing Unit:** ESP32 firmware programmed for signal computation, filtering, and wireless communication.
- **Visualization Interface:** A responsive web-based dashboard for real-time monitoring, control, and data display.
- **Storage Module:** Micro SD card used for local data storage and post-experimental analysis.

This integrated architecture ensures smooth communication between hardware and software subsystems, resulting in a reliable, user-friendly, and efficient thrust measurement platform.

4.3 DASHBOARD DESIGN

The dashboard is designed with a clean, dark-themed user interface to enhance readability and reduce visual strain during testing operations. It provides a well-organized layout divided into two main sections — the **Control Panel** and the **Data Visualization Panel** — ensuring both intuitive control and real-time feedback for the user.

4.3.1 Control Panel

The control panel serves as the user's main interaction interface for managing the thrust test operations. It allows users to set the **throttle percentage** manually or define **automated test sequences** with adjustable parameters such as **ramp time** and **log interval.**

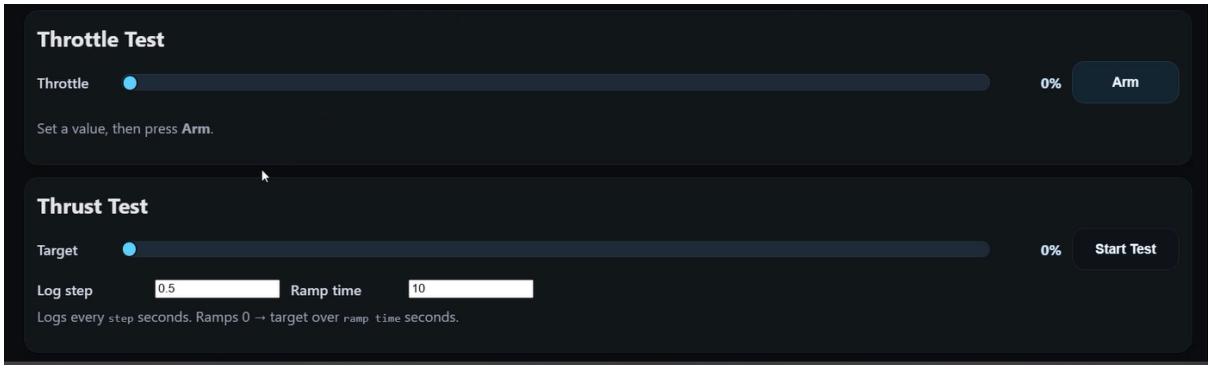


Figure 4.1 Control Panel

A dedicated **Arm/Disarm** feature ensures safe activation of the motor, preventing accidental start-up during setup or calibration. The panel also includes input fields for **target thrust levels**, enabling controlled and repeatable tests at specific power settings. The throttle slider provides smooth control over motor speed, allowing gradual acceleration to avoid sudden load variations on the system. This structured control mechanism ensures both safety and precision in the testing process.

4.3.2 Data Visualization Panel

The data visualization panel presents all live sensor data in the form of interactive time-series graphs, allowing users to monitor motor behavior instantly. Parameters such as **voltage (V)**, **current (A)**, **power (W)**, **thrust/load (kg)**, and **throttle (%)** are plotted dynamically as the test progresses. Each graph updates in real-time, offering clear insights into how motor performance varies with throttle input.

Figure 4.2 Data Visualization Panel

Additionally, the dashboard displays key setup information, including **motor brand, KV rating, propeller type**, and **battery configuration**, ensuring that all relevant test details are available on-screen for reference. The panel layout is responsive, providing a seamless experience on desktop and mobile devices alike. This visualization not only aids in quick decision-making during tests but also enhances the accuracy of post-experiment analysis by correlating electrical and mechanical parameters visually.

4.4 DASHBOARD FUNCTIONALITIES

The developed dashboard provides an integrated environment for real-time monitoring, control, and data analysis of the thrust test rig. It enables users to manage the entire testing process remotely through an intuitive and responsive web interface. The following are the major functionalities incorporated in the dashboard:

4.4.1 Throttle Control

The dashboard allows users to control the motor throttle using an interactive slider manually or automatically. The throttle value can be adjusted gradually to vary motor speed smoothly, preventing sudden changes in load that could affect measurement accuracy. The “Arm” and “Disarm” options ensure operational safety by preventing unintentional motor activation during setup or calibration.

4.4.2 Thrust Test Automation

In addition to manual control, the dashboard supports automated test sequences by allowing the user to define **ramp time**, **target throttle percentage**, and **logging intervals**. This functionality enables the motor to accelerate gradually to the target speed while the system records all relevant parameters at fixed time intervals. Automated testing improves consistency and repeatability across multiple trials, reducing manual effort and human error.

4.4.3 Real-Time Visualization

The dashboard continuously updates graphical plots of voltage, current, power, thrust (load), and throttle percentage in real-time. These live graphs provide immediate feedback on system performance and help in analyzing transient behaviours such as current spikes or thrust fluctuations during acceleration. This dynamic visualization assists in fine-tuning motor settings and understanding performance trends under different test conditions.

4.4.4 Data Logging

All test data, including time-stamped readings of electrical and mechanical parameters, are automatically stored on a micro-SD card connected to the ESP32. This feature ensures that the recorded data can be retrieved later for detailed post-processing, analysis, and report generation. The logged data supports

performance benchmarking and helps in identifying motor efficiency at varying loads and speeds.

4.4.5 Monitoring Parameters

The dashboard simultaneously displays all critical parameters — **voltage (V)**, **current (A)**, **power (W)**, **load (kg)**, and **throttle (%)** — providing a complete overview of the system's operating conditions. These synchronized readings enable accurate power-to-thrust and efficiency evaluations. The organized layout ensures clarity, allowing users to monitor and compare multiple variables effortlessly during live testing.

4.5 ADVANTAGES

The developed thrust measurement dashboard offers several advantages that enhance the accuracy, usability, and efficiency of the testing process. It provides a comprehensive environment for data acquisition, control, and visualization, reducing manual intervention and potential sources of error.

- **Real-Time and Remote Monitoring:**

  The dashboard enables live observation of all performance parameters through wireless connectivity, allowing users to monitor tests remotely from any device connected to the same network.

- **Intuitive Control Interface:**

  The simple and organized interface allows users to adjust throttle, start or stop tests, and configure test parameters without requiring complex coding or additional software tools.

- **Automated Testing and Data Logging:**

  The inclusion of automated test routines and built-in data storage eliminates the need for external software, ensuring seamless operation and continuous data capture throughout the test cycle.

- **Enhanced Accuracy and Repeatability:**

  By minimizing manual adjustments and introducing precise digital control, the dashboard ensures consistent test conditions, leading to more reliable and reproducible results.

- **Reduced Testing Time:**

  Automated throttle control, real-time data recording, and instant visualization significantly reduce the duration of testing and analysis, improving overall system productivity.

- **Comprehensive Performance Analysis:**

  The dashboard integrates electrical and mechanical parameters, enabling users to evaluate overall system efficiency and identify optimal motor operating points effectively.

4.6 REAL-TIME DATA DISPLAY

The real-time data display feature of the dashboard provides a clear and interactive visualization of all essential motor parameters. The dynamic plotting system continuously updates sensor readings such as **voltage**, **current**, **power**, **thrust**, and **throttle percentage** on time-based graphs. This allows users to observe the instantaneous response of the propulsion system during different operating conditions.

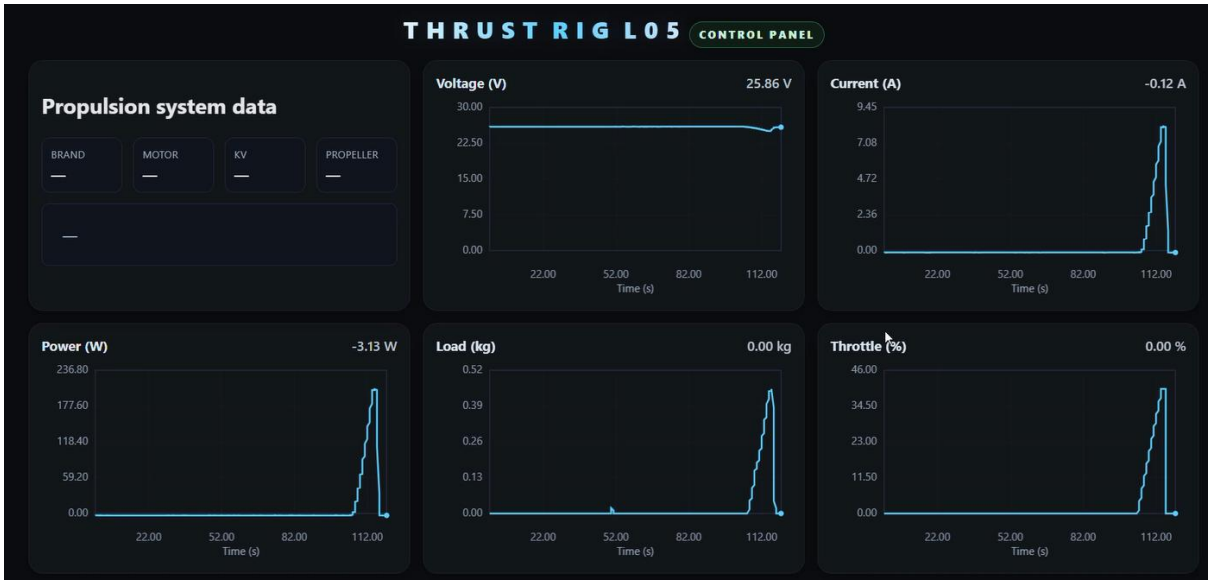


Figure 4.3 Real Time Data Visualization

The live graphs assist in detecting performance fluctuations, identifying inefficiencies, and validating theoretical predictions with experimental results. Voltage and current graphs indicate the electrical stability of the power supply, while power and thrust plots show the conversion efficiency of electrical energy into mechanical output. The throttle curve provides insights into the motor's response characteristics, enabling better understanding of its dynamic behaviour.

Additionally, the dashboard's smooth graphical interface ensures minimal delay between data acquisition and display, ensuring near real-time feedback. This functionality enhances user interaction during testing and helps in making quick adjustments if irregularities or anomalies are observed. Overall, the real-time visualization feature serves as a vital tool for performance evaluation, calibration, and validation of UAV propulsion systems.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 EXPERIMENTAL TEST RESULTS

Experimental testing was conducted on three propellers **5-inch**, **7-inch**, and **10-inch** using an **EMAX 1500 KV BLDC motor** mounted on the developed thrust measurement rig. The tests were performed at incremental throttle levels, and corresponding values of **current**, **power**, and **thrust** were recorded through the integrated sensors and dashboard system.

The tests were carried out under identical conditions with a fixed supply voltage and ambient environment to ensure consistency. Each propeller was tested individually to study the variation in motor load and performance characteristics with respect to propeller size.

5.5.1 Test Range and Data Collection

5-inch propeller

THROTTLE (in %)	VOLTAGE (in V)	CURRENT (in A)	THRUST (in kg)	POWER (in W)
10	16.836	0.259	0.029	4.325
20	16.313	1.302	0.093	21.241
30	15.971	2.41	0.152	38.495
40	15.637	3.499	0.207	54.708
50	15.156	4.981	0.262	75.488
60	14.799	5.96	0.301	88.196

Table 5.1 Experimental Results of 7-inch Propeller

7-inch propeller

THROTTLE (in %)	VOLTAGE (in V)	CURRENT (in A)	THRUST (in kg)	POWER (in W)
10	16.731	0.353	0.026	5.914
20	16.335	1.633	0.145	26.676
30	15.909	3.214	0.276	51.137
40	15.251	5.433	0.417	82.841
50	14.728	7.203	0.513	106.085
60	14.143	9.13	0.607	129.131

Table 5.2 Experimental Results of 7-inch Propeller

5-inch and 7-inch propellers: Tested from **0% to 60% throttle**, since both exhibited stable mechanical performance and safe current draw throughout this range.

10-inch propeller

THROTTLE (in %)	VOLTAGE (in V)	CURRENT (in A)	THRUST (in kg)	POWER (in W)
10	18.477	0.364	0.084	6.726
20	18.36	2.313	0.283	42.467
30	18.138	6.111	0.577	110.842
40	17.797	11.809	0.957	210.165

Table 5.3 Experimental Test Results of 10-inch Propeller

10-inch propeller: Tested only up to **40% throttle**, as higher speeds resulted in significant vibrations and thrust fluctuations due to larger aerodynamic loading.

The ESP32-based dashboard automatically logged the readings of current (A), voltage (V), and thrust (g) at each throttle level. Power was computed using the equation:

$$\boldsymbol{P} = \boldsymbol{V} \times \boldsymbol{I}$$

and efficiency was later derived from the ratio of thrust to power.

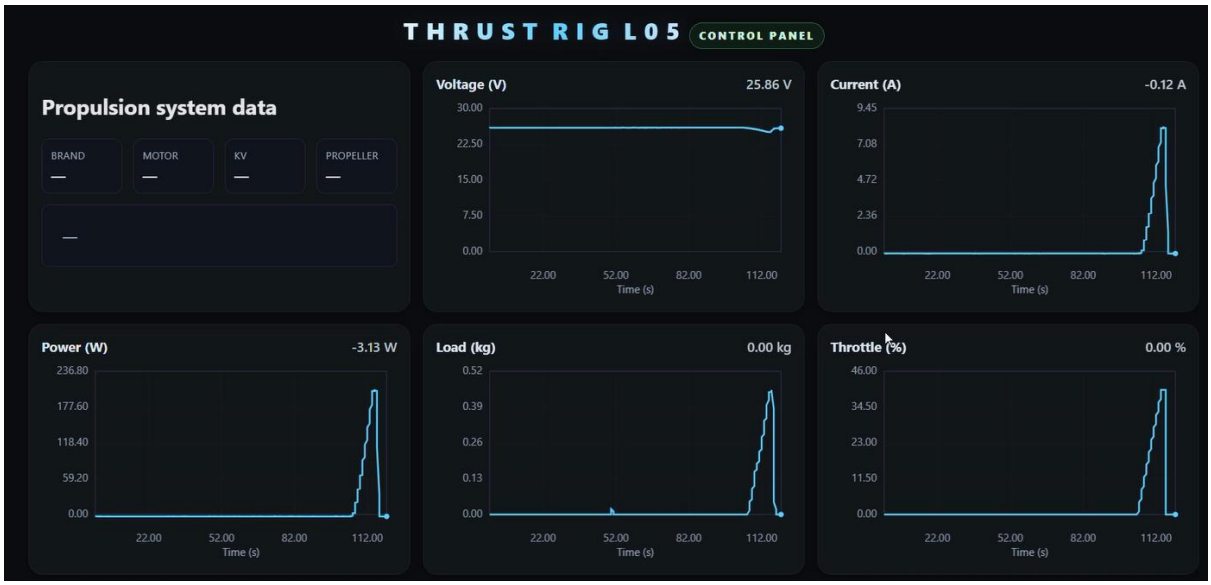


Figure 5.1 Dashboard Screenshot during Live Test

The observed trend showed that as throttle increased, **current and power** rose almost exponentially, while **thrust** increased linearly at lower levels and gradually saturated at higher throttle points. This pattern was consistent across all three propeller sizes but with varying magnitudes.

5.2 PERFORMANCE ANALYSIS

The experimental data collected from each propeller was analyzed to understand the relationship between **propeller diameter, motor load, and overall efficiency** of the propulsion system.

5.2.1 5-inch Propeller

The 5-inch propeller exhibited **lower thrust output** compared to the larger propellers but consumed minimal current, indicating low mechanical load on the motor. The thrust increased almost linearly up to 60% throttle, while current and power consumption grew steadily. This propeller configuration is suitable for **lightweight drones** or applications where **high maneuverability** and **lower power draw** are desired.

5.2.2 7-inch Propeller

The 7-inch propeller produced **moderately higher thrust** with a proportional increase in current. The power curve showed a steeper rise beyond 50% throttle, indicating the onset of aerodynamic drag. Despite higher consumption, the thrust-to-power ratio remained favourable, showing improved **propulsive efficiency** compared to the 5-inch propeller. This makes the 7-inch propeller an optimal choice for **medium-lift UAVs**, balancing thrust and efficiency effectively.

5.2.3 10-inch Propeller

The 10-inch propeller test, limited to **40% throttle**, showed the **highest thrust output** among all configurations but also a significant increase in current and power consumption. The data indicates that even at partial throttle, the 10-inch propeller imposes a **heavier load** on the motor due to higher aerodynamic drag and torque requirements.

The performance graphs for the 10-inch propeller showed:

- **Thrust vs Throttle:** Linear increase up to 40%, with smooth response and high static thrust generation.

Figure 5.2 Thrust vs Throttle Curve for 10-inch Propeller

- **Current vs Throttle:** Nonlinear increase, indicating exponential power demand at higher speeds.

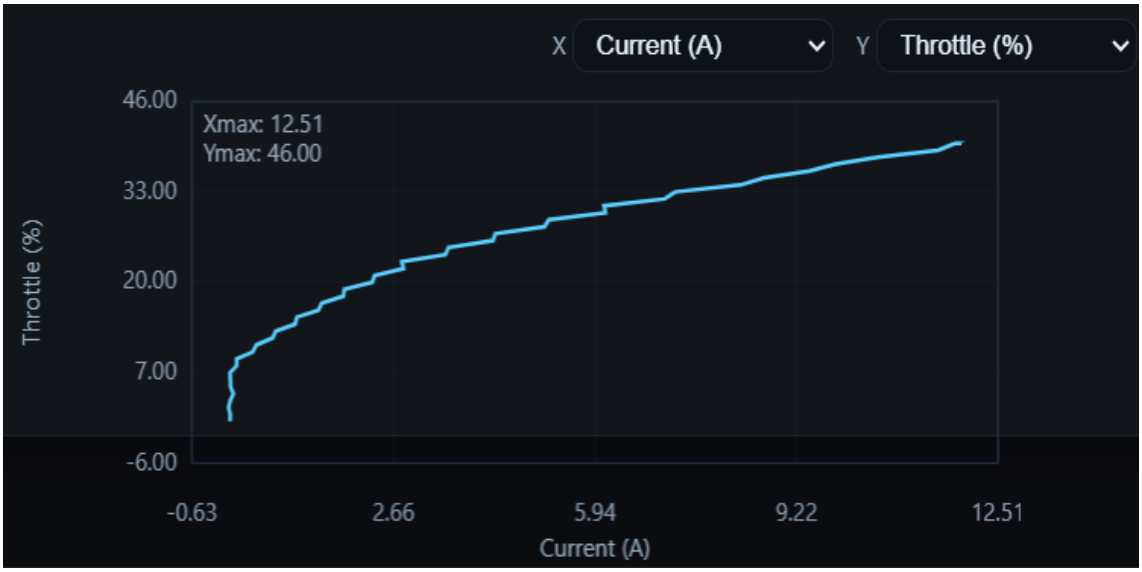


Figure 5.3 Current vs Throttle Curve for 10-inch Propeller

- **Power vs Thrust:** Quadratic relation, confirming higher energy input requirements for proportional thrust increments.

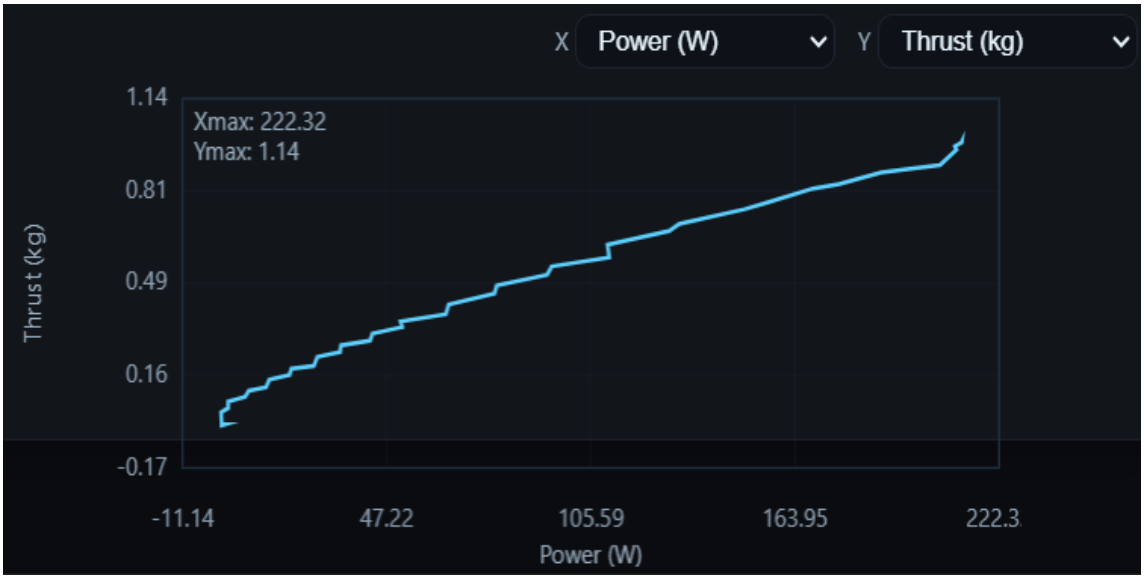


Figure 5.4 Power vs Thrust Plot for 10-inch Propeller

Although the 10-inch propeller achieved the highest static thrust, the efficiency dropped due to high current draw and vibration-induced losses at higher speeds. Therefore, this configuration is better suited for **heavy-lift UAVs** that prioritize force over endurance.

5.3 COMPARATIVE DISCUSSION ON PROPELLER PERFORMANCE

The comparative evaluation of the **5-inch, 7-inch, and 10-inch propellers** with the same **EMAX 1500 KV motor** provides insight into how propeller size influences thrust generation, current draw, and system efficiency.

5.3.1 Effect of Propeller Diameter

Larger propellers generate more thrust due to increased air displacement per revolution, but this comes at the cost of higher torque and current demand. The **5-inch propeller** required minimal power and produced the least thrust, while the **10-inch propeller** produced maximum thrust with significantly higher electrical input.

5.3.2 Efficiency and Power Loading

The **7-inch propeller** achieved the best balance between thrust and efficiency within the tested range. Its **thrust-to-power ratio** was highest at 40–50% throttle, indicating optimal operation where both aerodynamic and electrical parameters were well matched. The **10-inch propeller**, despite producing higher thrust, operated with reduced efficiency due to aerodynamic drag and high current draw.

5.3.3 Motor Behaviour

The EMAX 1500 KV motor demonstrated smooth performance across all tests. The motor showed **stable torque generation and minimal noise** within the tested throttle ranges, validating the proper calibration of the ESC and PWM control signals. The consistency of current-thrust data across all three propellers also confirms the accuracy of the measurement system.

Propeller Size	Throttle Range Tested	Max Thrust	Current Drawn	Efficiency	Suitable Application
5-inch	0 – 60%	Low	Low	Moderate	Micro or Racing Drones
7-inch	0 – 60%	Medium	Moderate	High (Optimal)	Medium UAVs/ Survey Platforms
10-inch	0 – 40%	High	High	Moderate to Low	Heavy-Lift / Payload UAVs

Table 5.4 Comparative Study on Motor-Propeller Performance

The results demonstrate that propeller size has a direct impact on UAV motor efficiency and performance characteristics. Smaller propellers favour efficiency and responsiveness, whereas larger ones provide greater lifting capacity but increase electrical power demands.

Overall, for the EMAX 1500 KV motor, the 7-inch propeller provides the best compromise between thrust, current draw, and efficiency, making it the optimal configuration for balanced UAV operations.

5.4 SUMMARY

This chapter presented the measured data and detailed performance analysis of the EMAX 1500 KV BLDC motor with three propeller sizes. The data validated the accuracy of the developed thrust measurement system and provided insights into how propeller size influences UAV performance.

Among all tested configurations, the 7-inch propeller achieved the highest efficiency within the tested throttle range, while the 10-inch propeller demonstrated superior thrust capability but lower energy efficiency. The results highlight the strong correlation between aerodynamic loading and electrical power demand, confirming that optimal UAV performance depends on careful propeller–motor matching.

These findings establish a reliable foundation for future UAV propulsion studies, including full-throttle performance testing, propeller optimization, and advanced motor efficiency modelling under dynamic flight conditions.

The experimental results also demonstrated the reliability and repeatability of the developed test rig, confirming its suitability for laboratory-based UAV propulsion evaluation. The observed variations in thrust and current across different propeller sizes align closely with theoretical aerodynamic models, validating the consistency of the measurement process.

Furthermore, the real-time dashboard enabled continuous monitoring of voltage, current, thrust, and efficiency, improving data accuracy and experimental control. The inclusion of external wind simulation provided a more realistic representation of in-flight aerodynamic effects, helping to identify performance deviations under varying load conditions.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The experimental evaluation of the **EMAX 1500 KV BLDC motor** using propellers of **5-inch, 7-inch, and 10-inch diameters** successfully demonstrated the working and effectiveness of the developed **thrust measurement test rig** and **real-time monitoring dashboard**. The integrated system accurately captured thrust, current, voltage, and power data across different throttle levels and provided real-time visualization for analysis.

The results validated that propeller size has a significant influence on motor load, efficiency, and thrust output. The **5-inch propeller** produced the least thrust but consumed minimal current, making it suitable for lightweight UAVs or racing drones. The **10-inch propeller** delivered the maximum thrust but also exhibited higher current draw and vibration, reducing overall efficiency. The **7-inch propeller**, however, provided a balanced performance with an optimal **thrust-to-power ratio**, achieving the highest efficiency within the tested throttle range.

The test rig performed reliably across all trials, confirming that the **load cell, ESC, and ESP32-based data acquisition system** functioned accurately and consistently. The results also highlighted the strong correlation between **thrust, power, and efficiency**, emphasizing the importance of selecting appropriate propeller–motor combinations to meet specific UAV mission requirements. Overall, the project successfully met its objectives of developing a **cost-effective, modular, and precise UAV thrust measurement setup** for academic and research use.

6.2 KEY FINDINGS ON OPTIMAL MOTOR PERFORMANCE

Based on the results obtained from the experiments, several key findings regarding the **optimal performance range** of the EMAX 1500 KV motor were observed:

1. **Optimal Throttle Range:**

   The motor operated most efficiently between **40% and 60% throttle**, where the increase in thrust was proportional to power consumption, and the system exhibited stable current draw without excessive heating.

2. **Propeller Efficiency Comparison:**

   - The **7-inch propeller** achieved the **best balance between thrust and efficiency**, producing sufficient lift for medium UAVs with minimal energy loss.
   - The **5-inch propeller** showed lower efficiency due to reduced aerodynamic loading, making it more suitable for smaller, high-speed drones.
   - The **10-inch propeller**, though capable of high thrust generation, suffered from efficiency loss and vibration effects at higher speeds, limiting its usefulness for sustained operation.

3. **Motor and System Performance:**

   The EMAX 1500 KV motor displayed consistent response to throttle inputs, indicating proper ESC calibration and PWM control from the ESP32. Current and power increased predictably with throttle, verifying the linearity of the control system.

4. **Performance Trade-offs:**

   Increasing propeller size enhances thrust but reduces efficiency due to higher aerodynamic drag and torque demand. This trade-off highlights the necessity of selecting propeller size according to UAV type — endurance, payload, or agility-focused.

These findings emphasize that for the **EMAX 1500 KV motor**, the **7-inch propeller configuration** offers the **optimal operating range**, delivering maximum performance efficiency and stable operation under moderate loads.

6.3 FUTURE SCOPE

The developed thrust measurement system performed effectively within the tested range. However, several enhancements can be incorporated to improve its efficiency, functionality, and research potential.

1. **Automation of Testing and Data Acquisition**

   The current setup requires manual throttle control and data supervision. Automating the entire testing and data acquisition process would significantly enhance both efficiency and accuracy. By implementing a **closed-loop control mechanism** with programmed throttle sequences, the system can perform multiple test cycles autonomously while maintaining consistent environmental and operational conditions. Automated data logging and analysis would also reduce human error, ensure uniform sampling intervals, and allow for long-duration endurance testing without manual intervention.

2. **Integration of Advanced Sensors**

   To expand the scope of performance evaluation, additional sensors such as **vibration, temperature, and acoustic sensors** can be integrated into the

system. Vibration data would help assess mechanical stability and propeller balancing, while noise measurements can be used to evaluate the acoustic efficiency of different propeller designs. Temperature sensors can monitor motor and ESC heating, enabling thermal efficiency analysis. These additional measurements would provide a more comprehensive understanding of UAV propulsion dynamics and component reliability under various load conditions.

3. **Cloud-Based Analytics and Remote Monitoring**

   Incorporating **cloud connectivity** into the system can transform it into an IoT-enabled testing platform. Experimental data can be uploaded to a secure cloud database in real time, allowing researchers to monitor tests remotely and analyze results from any location. Cloud-based analytics can facilitate **comparative performance studies** across different motor–propeller combinations, visualize trends using online dashboards, and support collaborative research between institutions. This enhancement would make the system scalable, user-friendly, and suitable for large datasets generated from repeated UAV propulsion tests.

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